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SEED: Self-Evolving On-Policy Distillation for Agentic Reinforcement Learning

Converting On-Policy Trajectories into Hindsight Skills and Distilling Them Back into the Policy Closes the Credit Assignment Gap in Long-Horizon Agentic Tasks

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Outcome-based reinforcement learning for long-horizon agentic tasks provides only a single reward signal at the end of trajectories spanning 20–50 decision steps, leaving a large credit assignment gap at the token level. SEED fills this gap by converting completed on-policy trajectories into natural-language hindsight skills—reusable workflows, decisive observations, failure-avoidance rules—and distilling them back into the same policy that generated the trajectories. The policy and its hindsight supervisor co-evolve across rounds, outperforming both pure RL and static distillation baselines on AlfWorld and WebArena.

AT A GLANCE

Problem: Sparse outcome rewards in long-horizon agentic RL provide insufficient token-level supervision, leaving the policy to learn intermediate decision quality from a single scalar.

Why it matters: Better intermediate supervision enables smaller models to learn complex agentic tasks—SEED yields the largest gains at 3B parameters.

Method: The policy analyzes its own completed trajectories into hindsight skills, then is distilled on (x, skill, a) triplets; the loop repeats as the policy evolves.

Result: AlfWorld success rate 74.2% (7B) vs. 72.8% GRPO-only; WebArena 26.3% vs. 18.4% GRPO-only.

Evidence: Fixed analyzer loses 5.1pp; scrambled skills fall below GRPO-only, confirming hindsight content drives gains.

The Big Picture

Training LLMs to act as autonomous agents—navigating web UIs, executing shell commands, managing files across multi-step workflows— requires learning from environmental feedback over trajectories that may span dozens of decisions. GRPO and similar outcome-based RL methods assign the same reward to every token in the trajectory based on the final outcome, providing no signal about which intermediate actions were decisive.

SEED treats the policy as its own supervisor: after completing a trajectory, the same model reads it and generates a structured retrospective capturing what worked, what failed, and what it should do differently—then trains on this signal. When the policy improves, its next-round retrospectives become more accurate, creating a self-evolving supervision signal that stays aligned with the model throughout training.

Why Existing Approaches Fall Short

Pure outcome RL (GRPO): Gradient propagates from the final reward backward through all steps. Effective action-level attribution requires the model to implicitly learn which actions contributed to the outcome—a difficult credit assignment problem over long trajectories.

Static distillation: A stronger model (teacher) generates demonstrations or skills, and the student is trained on them. The teacher is fixed: once the student surpasses the teacher on some tasks, the distilled supervision becomes stale and may even be harmful.

Off-policy hindsight: Learning from past trajectories (collected by earlier policy versions) introduces distribution shift between the generating policy and the current policy. Skills derived from an earlier policy's behavior may not apply to the current policy's failure modes.

Core Mechanism

SEED runs three operations using the same model checkpoint, synchronized in a single training round:

1. **Act:** Policy π_k rolls out trajectories $\{\tau_i\}$ in the environment.
2. **Analyze:** Policy π_k reads each completed τ_i and generates a natural-language hindsight skill s_i: a 2–4 sentence description of the key decision pattern, crucial observation, or failure cause in that trajectory.
3. **Distill:** Policy π_{k+1} is trained on (x_i, s_i, a_i) triplets where x_i is the task context, s_i is the hindsight skill, and a_i is the correct action.

After round k, π_{k+1} becomes the actor and analyzer for round $k + 1$. Skills generated by π_{k+1} reflect its own current capabilities and remaining failure modes.

Technical Formulation

Let $\tau = (o_1, a_1, \ldots, o_T, a_T, R)$ be a trajectory. The hindsight skill generation prompt $\mathcal{P}_{\text{analyze}}(\tau)$ instructs the model to identify: (a) which intermediate action most determined the outcome, (b) which environmental observation was the pivotal evidence, and (c) which alternative action would have changed the result.

The skill generation objective:

$$s_i = \pi_k(\cdot \mid \mathcal{P}_{\text{analyze}}(\tau_i))$$

The distillation objective (behavior cloning with skill context):

$$\mathcal{L}_{\text{SEED}}(\theta) = -\mathbb{E}_{(x,s,a)\sim\mathcal{D}_k}[\log \pi_\theta(a \mid x, s)]$$

The full training objective mixes outcome RL and skill distillation:

$$\mathcal{L}_{\text{total}} = (1-\alpha)\,\mathcal{L}_{\text{GRPO}} + \alpha\,\mathcal{L}_{\text{SEED}}$$

with $\alpha = 0.3$ in all experiments. GRPO provides the global policy gradient; SEED provides the local, token-level supervision.

Learning or Inference Procedure

Each SEED round proceeds as:

1. Sample B trajectories from π_k in parallel.
2. For each trajectory, run the analysis prompt to obtain s_i.
3. Construct the distillation dataset $\mathcal{D}_k$.
4. Train π_{k+1} on $\mathcal{L}_{\text{total}}$ for one epoch.

At inference, the model is queried without any hindsight skill context; skills are a training-time scaffold only, not an inference-time input.

What the Guarantee Says

SEED does not guarantee that hindsight skills are accurate—the model may generate incorrect retrospectives about its own behavior. What it does guarantee is *alignment*: since the analyzer and actor share weights, the analysis is always conducted from the perspective of the current policy's capabilities. As the policy improves, its analyses improve with it. This is a softer guarantee than formal credit assignment—but it holds without any external supervision.

Experimental Findings

AlfWorld (household task success ↑):

Method	3B	7B
GRPO only	61.3	72.8
Static distillation	67.4	78.1
SEED (ours)	**74.2**	**83.5**

WebArena (task success ↑):

Method	7B
GRPO only	18.4
Static distillation	21.7
SEED (ours)	**26.3**

The 3B model shows the largest proportional gain (+13pp on AlfWorld vs. GRPO-only), confirming that explicit hindsight supervision is most valuable when model capacity is limited.

Ablations and Interpretation

Fixed analyzer (π_0 as analyzer throughout): AlfWorld accuracy drops 5.1pp relative to full SEED. Stale analyzers produce skills that reflect initial failure modes, not current ones.

Scrambled skills (random pairing of skill s_j with context x_i): Accuracy falls to 59.8% on AlfWorld—below GRPO-only at 61.3%. Incorrect skills actively hurt performance, confirming that content quality, not training format, drives the improvement.

Analysis without distillation (GRPO + analysis as internal chain-of-thought): Improves AlfWorld by 1.4pp over GRPO-only. The self-analysis provides modest benefit as a form of implicit chain-of-thought, but the formal distillation step is responsible for the majority of gains.

α sensitivity: $\alpha \in [0.2, 0.4]$ gives similar results. $\alpha > 0.6$ degrades GRPO performance as distillation dominates.

Reference

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